



Development and application of an interactive generic physiologically based pharmacokinetic (igPBPK) model for adult beef cattle and lactating dairy cows to estimate tissue distribution and edible tissue and milk withdrawal intervals for per- and polyfluoroalkyl substances (PFAS)

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ABSTRACT

Humans can be exposed to per- and polyfluoroalkyl substances (PFAS) through dietary intake from milk and edible tissues from food animals. This study developed a physiologically based pharmacokinetic (PBPK) model to predict tissue and milk residues and estimate withdrawal intervals (WDIs) for multiple PFAS including PFOA, PFOS and PFHxS in beef cattle and lactating dairy cows. Results showed that model predictions were mostly within a two-fold factor of experimental data for plasma, tissues, and milk with an estimated coefficient of determination (R^2) of >0.95 . The predicted muscle WDIs for beef cattle were <1 day for PFOA, 449 days for PFOS, and 69 days for PFHxS, while the predicted milk WDIs in dairy cows were <1 day for PFOA, 1345 days for PFOS, and zero day for PFHxS following a high environmental exposure scenario (e.g., 49.3, 193, and 161 ng/kg/day for PFOA, PFOS, and PFHxS, respectively, for beef cattle for 2 years). The model was converted to a web-based interactive generic PBPK (igPBPK) platform to provide a user-friendly dashboard for predictions of tissue and milk WDIs for PFAS in cattle. This model serves as a foundation for extrapolation to other PFAS compounds to improve safety assessment of cattle-derived food products.

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) are a large class of more than 5,000 synthetic chemicals that have been utilized worldwide for decades. Due to their oil-, stain-, and water-repellent properties, PFAS have been used in a variety of consumer products and industrial applications globally (ATSDR, 2018). Because of their extensive use, PFAS have been detected in nearly every environmental medium on earth,

including soil, groundwater, as well as in animals and humans (Buck et al., 2011). The bioaccumulation, transfer, biomagnification and potential toxic effects of PFAS are of concern, primarily because PFAS compounds have a tendency to be mobile and persistent in the environment. Certain PFAS compounds such as perfluorooctanoic acid (PFOA), perfluorooctane sulfonate (PFOS) and perfluorohexane sulfonate (PFHxS) are capable of bioaccumulating in humans, which has potential to result in health issues in human population. Growing

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evidence has demonstrated that various adverse health outcomes, including impaired immune functions (Beans, 2021), developmental effects (Bach et al., 2016), liver damage (Costello et al., 2022) and cancer (Steenland and Winquist, 2021), have been linked to PFOA, PFOS and PFHxS exposure.

Edible food animal-derived products are potential sources of PFAS exposure for human population, largely because food animals can be exposed to PFAS from various contamination sources (EFSA et al., 2020; Lupton et al., 2022). For example, drinking water was polluted by industrial discharge and run-off from military bases where aqueous film forming foam containing PFAS has seeped into the soil and groundwater after their long use in extinguishing jet fuel fires (Milley et al., 2018; Young et al., 2021). Farms raising food animals can be contaminated by nearby factories and military installations, leading to a negative economic and environmental impact with potential ramifications on human and animal health (Jha et al., 2021). PFAS compounds have been found in products from several dairy farms in high dairy producing states in the U.S., including Maine, New Mexico, Michigan and Wisconsin (FDA, 2023). Based on the U.S. FDA report, PFAS compounds have been found in milk samples from dairy farms in New Mexico, including PFOA (0.047–0.169 ppb) and PFOS (0.881–5.680 ppb) (Cosier, 2019). Due to a lack of safe concentrations for PFAS in milk and edible tissues, farmers had to sacrifice farm animals and dump thousands of gallons of milk without any compensation (Honeycutt, 2022). This life and economic loss could have been prevented if a model for estimating tissue and milk residue depletion were available to estimate withdrawal intervals (WDIs) for PFAS-contaminated farm animal species after they are removed from PFAS contaminated sites. An ideal approach is the use of a physiologically based pharmacokinetic (PBPK) model developed and validated using data from PFAS contaminated farm animal species.

PBPK modeling is a mechanism-based computational approach that incorporates physiological and chemical-specific parameters to describe absorption, distribution, metabolism, and excretion of chemicals and their metabolites in different species (Chou and Lin, 2019; Chou et al., 2022; Li et al., 2019a; Tardiveau et al., 2022). Over the past few decades, PBPK models have been broadly used as a science-based approach for different applications, including chemical and nanoparticle risk assessment (Chen et al., 2022b; Lin et al., 2022; Tan et al., 2020), drug discovery and development (Naga et al., 2022; Rostami-Hodjegan and Toon, 2020; Zhuang and Lu, 2016), and animal-derived food safety assessment to support public health decision-making (Lautz et al., 2019; Li et al., 2017a; Lin et al., 2016; Zhou et al., 2021). Particularly, in risk assessment for PFAS compounds, several PBPK models have been developed in mice, rats, monkeys, and humans (Chen et al., 2022a; Cheng and Ng, 2017; Chou and Lin, 2019, 2021; Loccisano et al., 2013; Sweeney, 2022). The well-established toxicokinetic characteristics of PFAS compounds, which involves efficient absorption from the gastrointestinal tract, highly bound to plasma albumin, minimal metabolism, and primarily excreted via the kidney, has been a foundational consideration in previous PBPK models for PFAS (Chen et al., 2022a; Cheng and Ng, 2017; Chou and Lin, 2019, 2021; Loccisano et al., 2013; Sweeney, 2022). Due to the large variability in the plasma half-life of many PFAS compounds between species, prior PBPK modeling studies have considered the description of saturable renal reabsorption of PFOA and PFOS in the models based on *in vitro* studies of transporter activities in apical and basolateral proximal tubules to simulate the toxicokinetic behavior of PFAS compounds in rodents and humans (Chou and Lin, 2019; Loccisano et al., 2012a; Worley and Fisher, 2015; Worley et al., 2017). In terms of food-producing animals, a toxicokinetic model for PFAS compounds has been developed in laying hens (Kowalczyk et al., 2020), but PBPK models for PFAS compounds are not available in other food-producing animals, notably beef cattle and dairy cows which are important food animal species that have the potential to be contaminated in the U.S. Therefore, it is necessary to develop PBPK models for various PFAS compounds in different food animal species to estimate tissue residues, help determine safe concentrations in food products, and

predict WDIs for animal-derived food products.

Previously, our laboratory developed a comprehensive PBPK model for a predominant PFAS compound, i.e., PFOS in mice, rats, monkeys, and humans within a Bayesian framework (Chou and Lin, 2019) and then extrapolated it to other life stages such as gestation and lactation to support risk assessment of sensitive subpopulations, such as pregnant women, fetuses, and neonates (Chou and Lin, 2021). Based on our previous PFOS PBPK models and the scientific data gaps mentioned above, the objective of this study was to develop a generic PBPK model for selected PFAS compounds that can be used to simulate tissue and milk residues and estimate WDIs for multiple PFAS compounds in beef cattle and dairy cows. In this model, PFOA, PFOS and PFHxS were chosen as representative PFAS compounds because they are the most commonly detected PFAS compounds in milk and edible tissues in farm animals (Death et al., 2021). In addition, cattle (*Bos taurus*), including beef cattle and dairy cows, were selected as the model species because cattle-derived meat and milk are included in the FDA's list of major food in the U.S. (FDA, 2014). For our study, PFAS compound-specific tissue-to-plasma partition coefficients were determined/calculated based on published experimental data (Drew et al., 2019) and were used in a generic PFAS PBPK model, so that the model had flexibility to be extended to other PFAS compounds and other farm animal species. In order to ensure the model's user-friendliness, the final PBPK model was converted to a web-based visualization platform.

2. Materials and methods

2.1. Pharmacokinetic data for model development

All pharmacokinetic data used for model development were collected by searching on PubMed with the keywords: "Cattle", "Dairy Cows", and "PFAS". A summary of key information of selected pharmacokinetic studies is provided in Table 1. Time-dependent concentration profiles of selected PFAS compounds in plasma, tissues and milk were either extracted from tables or digitized from figures by using WebPlotDigitizer (version 4.4, <https://automeris.io/WebPlotDigitizer>) in the selected literature.

2.2. Generic PBPK model structure

A generic PBPK model for beef cattle and dairy cows (Fig. 1) was developed to predict the time-course pharmacokinetic profiles for selected PFAS compounds (i.e., PFOA, PFOS and PFHxS). The developed PFOS PBPK models for adult male rats and lactating female rats from previous studies (Chou and Lin, 2019, 2021) were used as a basis to extrapolate to the present model for beef cattle and dairy cows. The beef cattle model consisted of four compartments including liver, kidney, muscle, and rest of body. The model structure for dairy cows was consistent with the beef cattle model with the addition of udder and milk compartments. All organs or tissues were assumed to be homogenous compartments and connected by the circulating blood system (plasma compartment). The oral exposure route was incorporated into the model based on available experimental data and case study scenarios. Oral exposure to PFAS was described as a two-compartment model consisting of stomach (i.e., the entire bovine four-compartment stomach) and intestines to simulate the absorption of chemicals in the stomach and then transport to the intestinal tract via gastric emptying from the stomach (Equations S1–S3). Based on the prior studies, the processes of enterohepatic circulation and renal reabsorption play an important role in the PFAS fate and distribution (Death et al., 2021). Therefore, a physiologically based description of enterohepatic circulation and renal reabsorption with basolateral and apical transporters was considered in both beef cattle and dairy cow models (Equations S4–S6 for enterohepatic circulation and Equations S7–S10 for renal reabsorption). Elimination of selected PFAS compounds in the model were described through feces (Equation S2), bile (Equation S5), urine (Equation S11) and milk (dairy

Table 1

Summary of pharmacokinetic studies used for model calibration and evaluation of per- and polyfluoroalkyl substances (PFAS) compounds generic physiologically based pharmacokinetic (PBPK) model for adult beef cattle and dairy cows.

Reference	Routes	^a Dose (mg/kg)	^b Dosing regimen	Species/Breed	Age (month)	Matrix	Compounds	Use
Guruge et al. (2008)	PO	NA	NA	BC/Japanese Black	8–27	L, P	PFOS	C
Lupton et al. (2014)	PO	8	Single	BC/Australian Lowline	8–27	P, L, M, K, Sp, Lu	PFOS	C
Lupton et al. (2015)	PO	0.098, 9.01	Single	BC/American Angus	8–27	P, L, M, K	PFOS	C
Lupton et al. (2012)	PO	1	Single	BC/Australian Lowline	8–14	P, U, F	PFOA	C
Drew et al. (2019)	PO	0.00042	2 years	BC/Belted Galloway	8–18	P, L, M, K	PFOA, PFOS, PFHxS	C
Wang et al. (2017)	NA	NA	NA	BC/NA	NA	L, Milk	PFOA, PFOS	E
Kowalczyk et al. (2013)	PO	2, 7.6, 3.4, 4.6 (ug/L)	28 days	DC/Holstein Friesian	25	L, K, Milk, M, P, U	PFOA, PFOS, PFHxS	C
Vestergren et al. (2013)	PO	0.3 (ug/day)	NA	DC/Swedish Red	24	L, P, M, U, F, Milk	PFOA, PFOS	C
Xing et al. (2016)	NA	NA	NA	DC/NA	NA	Milk	PFOS, PFOA	E

Abbreviations: BC, Beef cattle; C, calibration; DC, dairy cows; E, evaluation; F, fat; K, kidneys; L, liver; Lu, Lung; M, Muscle; P, plasma; PO, oral administration; NA, not available or not applicable.

^a If the dosing information was not available in the literature; we calculated the daily PFAS intake by using Equations S16–S17 based on the PFAS levels measured from Maine EPA (MDEP, 2022) (Table S1).

^b If there was no information about exposure durations or dosing regimens, we assumed 2 years as exposure duration because the typical age range for butchering steers and entering milk production is around two years old.

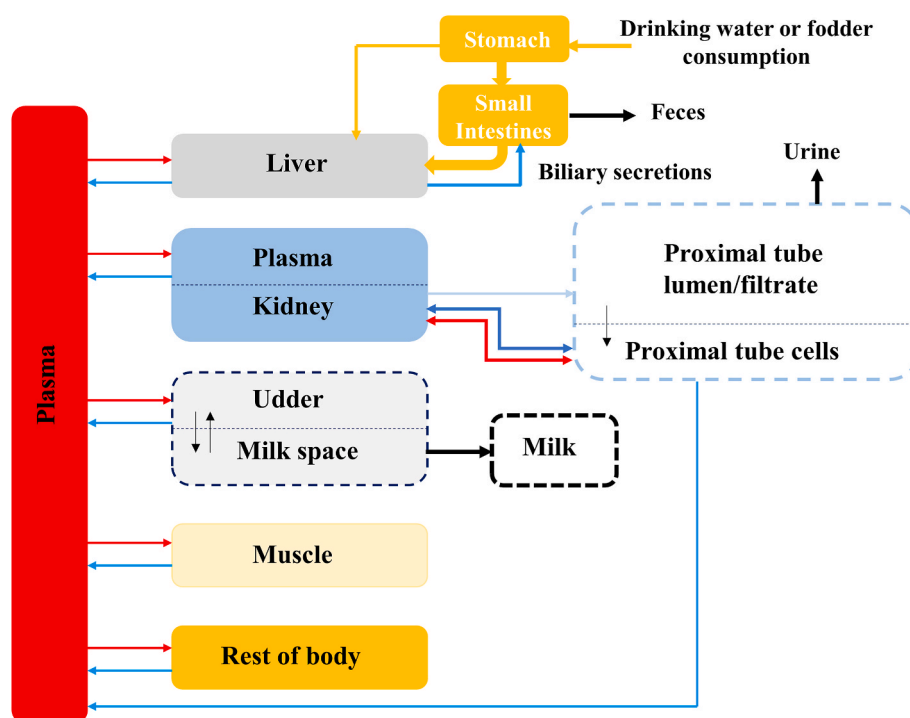


Fig. 1. A schematic of the generic PBPK model for PFOA, PFOS and PFHxS for beef cattle and dairy cows. The beef cattle model consisted of four compartments including liver, kidney, muscle, and rest of body. The model structure for dairy cows was consistent with the beef cattle model with additional udder and milk compartments.

cows only; Equations 12–13). Mass balance differential equations were used to describe the rate of change of PFAS compounds for each of the compartments defined in our model (Equations S14–15). In addition, since the selected PFAS compounds are very stable in the human body, metabolism of selected PFAS compounds was not considered, which is consistent with PBPK models for PFAS compounds in other mammalian species (Chou and Lin, 2019, 2021; Fabrega et al., 2014; Loccisano et al., 2011, 2012a).

2.3. Model parameterization and calibration

Parameters for the generic PBPK model for PFAS compounds consist of species-specific physiological parameters and chemical-specific parameters. Species-specific physiological parameters, including body weight (BW), cardiac output (QCC), fractions of blood flow to individual tissues (e.g., the fraction of blood flow to liver, QLC) as well as the

volume fractions of individual organs (e.g., the volume fraction of liver, VLC), were collected from a previous review article (Lin et al., 2020) that summarized published experimental data. All species-specific physiological parameters used in this model are detailed in Table 2. Chemical-specific parameters included protein binding parameters, absorption/elimination rate constants, partition coefficients, enterohepatic circulation and renal reabsorption parameters. Due to a lack of experimental values, protein binding parameters (free), absorption (KabsC) and elimination rate constants (KfecesC, KurineC and KbileC), and renal reabsorption parameters (Vmax_baso_invitro, Km_baso, Vmax_apical_invitro and Km_apical) for beef cattle and dairy cow models were assumed to be the same as fitting values from previously published PFOS PBPK models in rats (Chou and Lin, 2019). The rate constant of enterohepatic circulation (KehcC) for cattle and cows was obtained from the optimized values from a previously published generic PBPK model in cattle (Chou et al., 2022). Tissue-to-plasma partition

Table 2

Physiological parameters values used in the generic per- and polyfluoroalkyl substances (PFAS) compounds physiologically based pharmacokinetic (PBPK) model for adult beef cattle and dairy cows.

Parameter	Abbreviation	Beef Cattle	Dairy Cows
Cardiac output (L/h/kg)	QCC	5.45 (1.47)	5.45 (1.47)
Hematocrit	Htc	0.378 (0.046)	0.29 (0.087)
Glomerular filtration rate constant (L/h/kg of kidney)	GFRC	0.138	0.138
Blood flow (fraction of cardiac output, unitless)			
Liver	QLC	0.44 (0.25)	0.58 (0.19)
Kidney	QKC	0.11 (0.08)	0.07 (0.10)
Muscle	QMC	0.28 (0.09)	0.18 (0.054)
Udder	QUC	–	0.13 (0.06)
Rest of body	QRestC	1-QLC-QKC-QMC-(QUC) ^a	
Tissue volume (fraction of body weight, unitless)			
Plasma	VPC	0.0399 (0.0068)	0.0433 (0.0088)
Liver	VLC	0.0122 (0.0018)	0.0135 (0.0028)
Kidney	VKC	0.0021 (0.0005)	0.0022 (0.0005)
Muscle	VMC	0.361 (0.1173)	0.27 (0.081)
Udder	VUC	–	0.0314 (0.0050) ^b
Rest of body for metabolites	VRestC	1-VPC-VLC-VKC-VMC-(VUC) ^a	

Note: All parameter values in both beef cattle and dairy cows were collected from Lin et al. (2020), except QMC in beef cattle and dairy cows and QUC in cows were collected from Li et al. (2017b). The value was expressed as “mean (SD)”.

^a QRestC was calculated as (1-QLC-QKC-QMC) in beef cattle and (1-QLC-QKC-QMC-QUC) in dairy cows, and VRestC was calculated as (1-VPC-VLC-VKC-VMC) in beef cattle and (1-VPC-VLC-VKC-VMC-VUC) in dairy cows.

^b VUC was collected from the tissue volume of mammary gland from lactating cows in Lin et al. (2020).

coefficient parameters were collected from experimental data in cattle (Drew et al., 2019) and used in both beef cattle and dairy cow models. The parameters related to the udder compartment in dairy cows, including milk-to-mammary gland partition coefficient (PMilkM) and milk suckling or excretion rate constant (Kmilking), were obtained from a previously published lactational PFOS PBPK model in rats (Chou and Lin, 2021). These abovementioned parameter values were set as initial values in the model calibration. A preliminary sensitivity analysis was performed to identify highly sensitive parameters that were then optimized with measured pharmacokinetic data (i.e., training datasets) using Levenberg-Marquardt least-squares algorithm implemented with modFit function in the R package FME (Soetaert and Petzoldt, 2010). Briefly, by comparing the predicted values $f(t_{ij}, \theta)$ with the mean measured values at time point i of the data set j , the selected parameters θ were calibrated by least-squares function:

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \sum_{ij}^n \left[\frac{f(t_{ij}, \theta) - y_{ij}}{w_{ij} \times n_j} \right]^2$$

Where the w_{ij} is a weighting factor that normalizes the differences of units and magnitudes from different data sources. The w_{ij} was estimated from the standard deviation of measured values. N represents the number of data points for data set j which was used to scale the residuals to prevent abundant data set dominating the analysis. The model was fitted with a variety of pharmacokinetic datasets simultaneously and estimated the optimized parameters $\hat{\theta}$ at the minimum of sum of squared residuals for all data points in data sets. All collected and optimized chemical-specific parameters are provided in Table 3 for beef cattle and Table 4 for dairy cows.

Table 3

Chemical-specific parameter values used in the adult beef cattle physiologically based pharmacokinetic (PBPK) model for perfluorooctanoic acid (PFOA), perfluorooctane sulfonate (PFOS) and perfluorohexane sulfonate (PFHxS).

Parameter	Symbol	PFOA	PFOS	PFHxS
Absorption rate constant (/h)				
Rate of absorption in small intestines	<i>KabsC</i>	2.12	2.12	2.12
Tissue-to-plasma partition coefficient (unitless)^a				
Liver	<i>PL</i>	1.03	1.03	0.21
Kidney	<i>PK</i>	0.48	0.48	0.27
Muscle	<i>PM</i>	0.08	0.047	0.062
Rest of body	<i>PRest</i>	0.089	0.13	0.1
Free fraction of PFOS in plasma^b	<i>Free</i>	0.016	0.01	0.018
Rate constant for enterohepatic circulation (/h/kg)^c	<i>KehcC</i>	0.01	0.015	0.01
Transporter rates constant^b				
Vmax of basolateral (pmol/mg protein/min)	<i>Vmax_baso_invitro</i>	60.99	60.99	60.99
Km of basolateral (mg/L)	<i>Km_baso</i>	34.5	34.5	34.5
Vmax of apical (pmol/mg protein/min)	<i>Vmax_apical_invitro</i>	0.90	51803	9.77e4
Km of apical transporters (mg/L)	<i>Km_apical</i>	64.4	775	64.4
Relative activity factor^b				
Apical transporters (unitless)	<i>RAF_apical</i>	1e-6	8.99e-4	1.7e-4
Basolateral transporters (unitless)	<i>RAF_baso</i>	0.005	0.001	0.005
Eliminated and other rate constants (/h/kg BW^{0.25})^b				
Unabsorbed dose to appear in feces	<i>KfecesC</i>	0.0137	0.01	0.0137
Urinary elimination rate	<i>KurineC</i>	0.37	8.41e-4	0.1
Biliary elimination rate	<i>KbileC</i>	0.01	0.016	0.1
Rate of efflux of PFOS from PTCs into blood	<i>KeffluxC</i>	0.1	0.1	0.1
Diffusion rate from PTCs	<i>Kdif</i>	0.001	0.001	0.001

Note: The bold numbers represent the calibrated values that were fitted with experiment data (Table 1) using the Levenberg-Marquardt algorithm.

^a The baseline values of partition coefficients in beef cattle and dairy cows were collected based on measured data from Drew et al. (2019).

^b The baseline values were collected based on our previous PBPK model for PFOS in rats (Chou and Lin, 2019).

^c The baseline values were collected based on our previous PBPK model in cattle (Chou and Lin, 2021).

2.4. Model evaluation

The calibrated model was evaluated by comparing predicted values with observed values from the evaluation datasets (i.e., experimental pharmacokinetic data not included in the model calibration) (Table 1). The model performance was evaluated with statistical criteria and graphical goodness-of-fit following the PBPK model guidance from the World Health Organization (WHO) (WHO, 2010). The goodness-of-fit was analyzed, and the determination coefficient (R^2) was calculated based on linear regression between log-transformed observed and predicted values for both the calibration and evaluation results. The model was considered reasonable and acceptable when the simulated kinetic profiles were consistent with the observed pharmacokinetic profiles and predictions fell within a factor of 2 of the experimental data. The R^2 value was used as an indicator for the overall model prediction performance, and the model was considered acceptable if the R^2 value was equal to or higher than 0.75 (Li et al., 2018).

2.5. Sensitivity analysis

To identify sensitive parameters that have high influence on the model simulation (e.g., selected dose metrics), a local sensitivity analysis was performed to determine which model parameters had high

Table 4

Chemical-specific parameter values used in the dairy cows physiologically based pharmacokinetic (PBPK) model for perfluorooctanoic acid (PFOA), perfluorooctane sulfonate (PFOS) and perfluorohexane sulfonate (PFHxS).

Parameter	Symbol	PFOA	PFOS	PFHxS
Absorption rate constant (/h)				
Rate of absorption in small intestines	<i>KabsC</i>	2.12	2.12	2.12
Tissue-to-plasma partition coefficient (unitless) ^a				
Liver	<i>PL</i>	1.2	1.65	0.20
Kidney	<i>PK</i>	1.5	0.54	0.40
Muscle	<i>PM</i>	0.77	0.04	0.05
Udder	<i>PU</i>	0.2	0.2	0.20
Rest of body	<i>PRest</i>	0.05	0.08	0.10
Free fraction of PFOS in plasma	<i>Free</i>	0.013	0.015	0.04
Rate constant for enterohepatic circulation (/h/kg)	<i>KehcC</i>	0.002	0.014	0.010
Milk transfer and clearance parameters (unitless)				
Milk-to-mammary gland partition coefficient	<i>PMilkM</i>	0.958	0.958	0.176
Milk suckling rate constant	<i>Kmilking</i>	0.58	0.58	0.58
Transporter rates constant				
Vmax of basolateral (pmol/mg protein/min)	<i>Vmax_baso_invitro</i>	0.40	20	0.04
Km of basolateral (mg/L)	<i>Km_baso</i>	27	34.5	27
Vmax of apical (pmol/mg protein/min)	<i>Vmax_apical_invitro</i>	0.945	5.18e4	0.947
Km of apical transporters (mg/L)	<i>Km_apical</i>	52.3	157	243
Relative activity factor				
Apical transporters (unitless)	<i>RAF_apical</i>	0.0005	0.01	50
Basolateral transporters (unitless)	<i>RAF_baso</i>	4.07	0.50	4.07
Eliminated and other rate constants (/h/kg BW ^{0.25})				
Unabsorbed dose to appear in feces	<i>KfecesC</i>	0.065	0.002	0.01
Urinary elimination rate	<i>KurineC</i>	0.50	0.01	0.476
Biliary elimination rate	<i>KbileC</i>	0.065	0.003	0.009
Rate of efflux of PFOS from PTCs into blood	<i>KeffluxC</i>	0.017	0.5	1.03
Diffusion rate from PTCs	<i>Kdif</i>	0.10	0.13	0.013

Note: The bold numbers represent the calibrated values that were fitted with experiment data (Table 1) using the Levenberg-Marquardt algorithm.

^bThe baseline values were collected based on our previous PBPK model for PFOS in rats (Chou and Lin, 2019).

^cThe baseline values were collected based on our previous PBPK model in lactating rats (Chou and Lin, 2021).

^a The baseline values of partition coefficients in beef cattle and dairy cows were collected based on measured data from Drew et al. (2019).

impacts on the 2-year area under the curves (AUCs) of muscle concentrations in beef cattle and milk concentrations in dairy cows based on real-world exposure scenarios. Normalized sensitivity coefficients (NSCs) were calculated by increasing 1% of model parameter values to evaluate the relative sensitivity of each parameter on the corresponding AUCs for each of the selected PFAS compounds using the mathematical equation previously reported (Chou et al., 2022; Lin et al., 2011). The detailed equation is listed in Supplementary Material (Equation S18).

2.6. Establishment of a population PBPK model and its application on case studies

2.6.1. Establishment of a population PBPK model

The population-based simulations were generated by an integration of Monte Carlo simulations with the generic PBPK model based on random sampling from designated probabilistic distributions for each parameter. All physiological and chemical-specific parameters were randomly sampled from specified distributions described by the means and coefficients of variances (CVs). Based on the assumptions from

previous studies (Chou et al., 2022; Li et al., 2017a), all physiological (with the exception of blood flow fractions) and chemical-specific parameters were defined as normal and log-normal distributions, respectively. Because of the high variability of blood flow fraction in beef cattle and dairy cows collected from experimental data (Lin et al., 2020), the log-normal distribution was defined to prevent the generation of negative values. The CVs for most of the physiological parameters including the BW, cardiac output, and tissue volume fraction of liver, kidneys and the fraction of blood flows in liver were estimated based on observed data from the review paper compilation (Lin et al., 2020). For other physiological parameters with unknown CV values, a CV of 30% was used as the default value. For chemical-specific parameters, the CV values of tissue-to-plasma partition coefficients, absorption and elimination rate constants, and protein binding constants were assigned as 20%, 30% and 10%, respectively, based on previous studies (Chou et al., 2022; Henri et al., 2017; Li et al., 2017a; Riad et al., 2021). The distributions for all parameters were symmetrically truncated (i.e., 2.5th and 97.5th percentiles were used as upper and lower bounds) to avoid the selection of extreme outliers. Additionally, to avoid unbalance in the model, all physiological parameters were adjusted to ensure that the sum of tissue volumes or fractions of blood flows were equal to 1 after random sampling from defined distributions.

2.6.2. Estimation of PFAS exposure in beef cattle and dairy cows

Based on real-world exposure scenarios, farm animals were assumed to be exposed to PFOA, PFOS and PFHxS via forage and water consumption at PFAS compound-contaminated farms. For exposure durations, we assumed that beef cattle and dairy cows were exposed to PFAS compounds for 2 years. This is because the age at slaughter of beef cattle typically ranges from 18 to 24 months (USDA, 2001), while the milk production cycle of a dairy cow typically begins at approximately 25 months of age (USDA, 2006). The PFAS exposure doses for farmed animals were calculated based on the water and forage PFAS concentrations using multimedia exposure equations (Equations S16–S17). Site specific soil and water PFOA, PFOS and PFHxS concentrations were extracted from a database developed by the Maine Department of Environmental Protection (MDEP) (MDEP, 2022), and the 5th, 50th and 95th percentile soil and water PFAS concentrations (Table S1) were used to represent PFAS exposure levels in the low-, medium and high-exposure scenarios, respectively. Forage PFAS concentration was estimated as the product of PFAS soil concentration and the soil-to-plant transfer factor (Table S2). Beef cattle and dairy cow PFAS compound exposures were estimated as the product of the PFAS compound concentration in water or forage and cattle/cow water or forage consumption rates (Tables S2–S3). Total daily PFAS compound exposure (mg/kg/day) in beef cattle or dairy cows was estimated as the sum of the exposure from forage and water (Table S3). The daily PFAS exposure and exposure duration were considered as dosing parameters in our PBPK model.

2.6.3. Derivation of action levels for PFAS

Currently there are no action levels (equivalent to safe concentrations) or tolerances (maximum residue limits [MRLs] in other countries) for PFOA, PFOS and PFHxS in edible tissues from beef cattle nor in the milk from dairy cows. This study followed the standard risk assessment method based on the report from the Maine Center for Disease Control & Prevention (CDC) (Maine CDC, 2017a, b) to calculate action levels for PFAS compounds. The calculated action level was considered as “calculated safe concentration” or “operational tolerance” that was used to calculate WDIs in this study. The action level was calculated using the following equation:

$$PFAS \text{ action level} = \frac{\text{Reference dose}}{\text{Edible tissue or milk consumption}} \times RSC$$

where the reference doses (RfDs) for PFOA (20 ng/kg/day) and PFOS

(20 ng/kg/day) derived from the U.S. EPA were used (EPA, 2016a, b). There is no RfD for PFHxS derived from the U.S. EPA. We adopted the RfD values for PFHxS derived from the New Hampshire Department of Environmental Services (4 ng/kg/day) (New Hampshire DES, 2019) because it is more conservative when comparing RfDs derived from other states (Smith, 2020). The denominator in this equation is the calculated daily intake value of edible tissues (e.g., muscle, liver, and kidney) and milk on a body weight basis (g/kg/day or L/kg/day). The consumption rate for muscle and milk was obtained from a document about the derivation of action levels for beef and milk from the Maine CDC (Maine CDC, 2017a, b). The Maine CDC calculated the daily beef and milk consumption in the U.S. general population based on the data from U.S. Department of Agriculture (USDA) Food Patterns Equivalents Database (FPED) and U.S. Centers for Disease Control and Prevention's (U.S. CDC) National Health and Nutrition Examination Survey (NHANES) (USCDC, 2020; USDA, 2019). The consumption rate for liver was assumed to be 100 g per week based on the recommended value of beef liver consumption (Kicinska et al., 2019), while the daily intake of kidney was calculated from the ratio of kidney weight to muscle weight and multiplying the muscle consumption rate due to lack of appropriate data. In this equation, the relative source contribution (RSC) factor was used to consider the proportion of the total exposure dose via selected exposure routes versus the dose from other exposure sources. However, due to lack of appropriate data to derive the value, the default value of 0.8 was used for selected PFAS compounds (Maine CDC, 2017a, b). Calculated parameters (e.g., consumption rate of edible tissues and milk) and estimated action levels (i.e., operational tolerances) of PFAS compounds for edible tissue or milk consumption are listed in Table S4.

2.7. Web-based applications

The PBPK model code was programmed using the R package “mrgsolve” (Elmokadem et al., 2019). Ordinary differential equations (ODEs) were solved by the DLSODA solver that was written in FORTRAN and therefore speed up the calculations. The final generic PFAS PBPK model in beef cattle and dairy cows was converted to a web-based application with the R package “Shiny” (Chang et al., 2023) based on recently published methods (Chou et al., 2022; Li et al., 2019b; Riad et al., 2021; Yuan et al., 2022).

2.8. Code availability and result reproducibility

Model source code and all datasets that were used to calibrate and evaluate the present PBPK model can be downloaded from GitHub using this link: <https://github.com/UFPBPK/FARAD-PFAS>. These data sets and source code files will allow readers to reproduce our results and apply our PBPK model for further research.

3. Results

3.1. Model calibration

The generic PBPK model for PFAS compounds in beef cattle and dairy cows was used to simulate PFOA, PFOS and PFHxS concentrations in edible tissues and milk; with the model then calibrated with observed data following different exposure scenarios according to the collected pharmacokinetics studies (Table 1). Overall, the model predictions agreed with these observed data through a global evaluation of goodness-of-fit. The coefficient of determination (R^2) of 0.93, 0.98 and 0.96 for PFOA, PFOS and PFHxS, respectively, in the beef cattle model were estimated by comparing with experimental datasets, whereas the R^2 values were 0.91, 0.98 and 0.99 for PFOA, PFOS and PFHxS, respectively, in the dairy cow model (Fig. 2A). Most model predictions were within a two-fold error factor of measured data in either beef cattle or dairy cow models (Fig. 2B). The percentage of the predictions for all PFAS compounds within 2-fold errors was 85% and 82% in beef cattle and dairy cows, respectively (Fig. 2B). In addition, the simulated results from the calibrated PBPK model visually matched with the observed kinetic profiles of PFAS in edible tissues and milk in beef cattle (Fig. 3) and dairy cows (Fig. 4).

The predicted time-course concentration for PFOS in muscle (Fig. 3A), plasma (Fig. 3B) and urine (Fig. 3C) were consistent with the experimental data from the study of Lowline Angus steers given a single 8 mg/kg body weight oral dose of PFOS (Lupton et al., 2014), while the predictions for PFOS concentrations in liver [Fig. 3D (9.1 mg/kg)] and muscle [Fig. 3E (0.098 mg/kg) and Fig. 3F (9.1 mg/kg)] were adequately correlated with the data from the study of steers and heifers following a single oral bolus dose of PFOS at 0.098 mg/kg or 9.1 mg/kg (Lupton et al., 2015). Other studies from Australia reported accumulation of PFAS residues in beef cattle on a farm where stock water was contaminated by PFAS compounds (Drew et al., 2019, 2021). Our beef cattle model was used to predict PFOA concentrations in plasma (Fig. 3G), PFHxS in muscle (Fig. 3H) and plasma (Fig. 3I), and PFOS in liver (Fig. 3J), muscle (Fig. 3K) and plasma (Fig. 3L) following estimated daily intake of PFOA (1 mg/kg/day), PFOS (4.2e-4 mg/kg/day) and PFHxS (8.4e-4 mg/kg/day) for 2 years based on these studies (Drew et al., 2019, 2021; Lupton et al., 2012). Model simulations (Fig. 3G–L) corresponded with the experimental data well.

For the dairy cow PBPK model, measured time-course PFAS concentration data were collected from the study of Holstein cows exposed to a PFAS-contaminated feed (the daily intake of PFOA, PFOS and PFHxS was 2, 7.6 and 4.6 $\mu\text{g/kg}$, respectively) for 28 days in which the plasma, liver, kidney, and muscle tissue, urine, and milk were sampled and analyzed (Kowalczyk et al., 2013). The developed dairy cow model was used to predict the concentrations of PFOS in kidney (Fig. 4A), liver

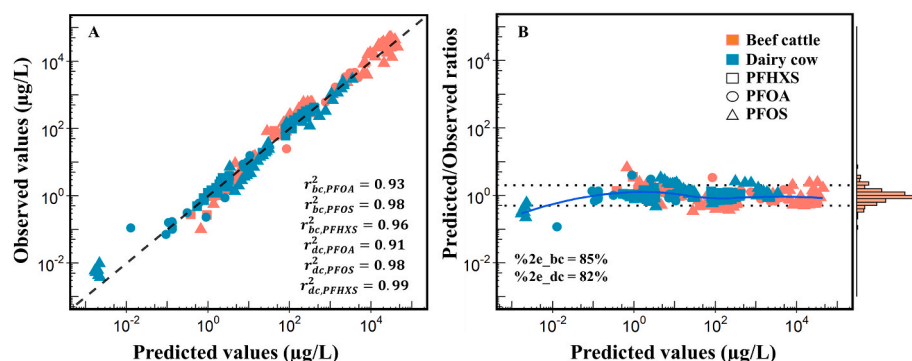


Fig. 2. Overall model calibration and evaluation results. (A) A global evaluation of goodness of model fit between the observed data (y-axis) and predicted values (x-axis), and (B) the predicted-to-observed ratios versus model prediction plot for PFOA, PFOS and PFHxS in beef cattle and dairy cows. The abbreviations r^2 represent the determination coefficients estimated based on calibration data and evaluation data sets. The abbreviations bc and dc represent the beef cattle and dairy cows, respectively. The %2e indicate that the percentage of predicted values was within 2-fold error range.

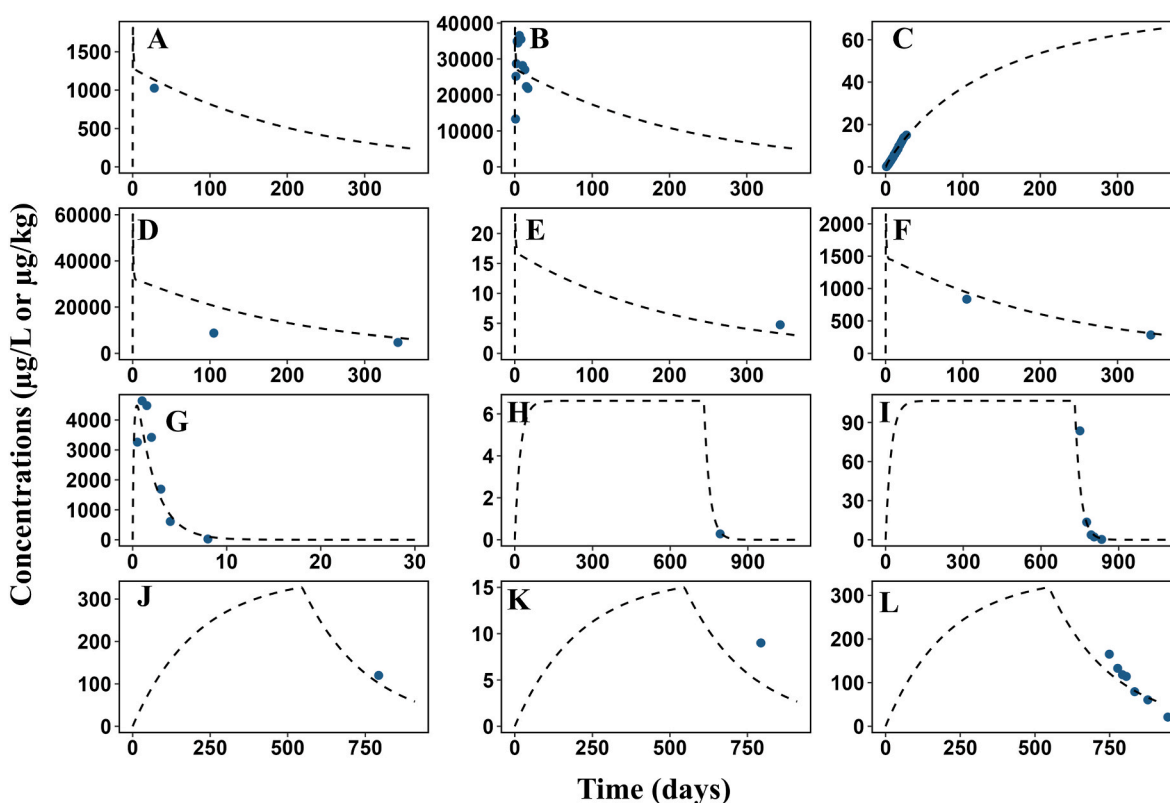


Fig. 3. Model calibration results for individual datasets of the PBPK model for PFOA, PFOS and PFHxS in beef cattle. Comparisons of model predictions (dashed lines) and observed data (blue circles) are shown for concentrations of PFOA, PFOS and PFHxS in the plasma, liver, and muscle from beef cattle. Experimental data are from the studies of (A–C) (Lupton et al., 2014), (D–F) (Lupton et al., 2015), (G) (Lupton et al., 2012) and (H–L) (Drew et al., 2019, 2021). Plots (A, B, C) represent PFOS concentrations in muscle, plasma, and urine, respectively; (D and E–F) represent PFOS concentrations in liver and muscle, respectively; (G) represents PFOA concentrations in plasma; (H, I) represent PFHxS concentrations in muscle and plasma, respectively; (J, K, L) represent PFOS concentrations in liver, muscle and plasma, respectively. Refer to Table 1 for the detailed dosing regimens from respective studies. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

(Fig. 4B) and milk (Fig. 4C), concentrations of PFOA in kidney (Fig. 4D), liver (Fig. 4E), muscle (Fig. 4F) and plasma (Fig. 4G), and concentrations of PFHxS in kidney (Fig. 4H), liver (Fig. 4I), milk (Fig. 4J), muscle (Fig. 4K) and plasma (Fig. 4L). Model simulations were in line with observed pharmacokinetic profiles.

3.2. Model evaluation with independent data

To further validate the model's predictability, this model was evaluated with an independent dataset that was not included in the model calibration. The model simulations were compared against measured data for PFOA and PFOS in muscle of beef cattle and milk of dairy cows (Table 1). Two studies that analyzed PFOA and PFOS concentrations in the beef muscle and liver samples (Wang et al., 2017) and milk in the Chinese market (Xing et al., 2016) were used as the evaluation dataset. Because there is no information on exposure scenarios, we estimated PFOA and PFOS exposure doses by using multimedia exposure equations (Equations S16–17) based on the measured PFOA and PFOS levels in water and soil samples in China (Li et al., 2019c, 2020). The exposure durations were assumed to be 2 years as defined above (Section 2.6.2). The results showed that the simulated levels correlated with the observed concentrations of PFOA and PFOS in edible tissues and milk (the ratios between predicted and observed values were generally within the two-fold error range), except for PFOA in milk and liver, and especially PFOS in muscle (Fig. 5). The model underestimated PFOA concentration in the milk (Fig. 5A) and liver (Fig. 5B), but the model overpredicted PFOS concentration in muscle (Fig. 5A).

3.3. Sensitivity analysis

Twenty-nine parameters were included in the local sensitivity analysis based on the dose metrics of 2-year AUCs of liver, kidney, muscle, and milk concentrations of PFOA, PFOS and PFHxS in beef cattle and dairy cows following real-world exposure scenarios (Fig. 6 and Supplementary Tables S5–S6). Fig. 6 is a representative figure showing the absolute percentage of NSCs based on 2-year AUCs of PFOA, PFOS and PFHxS for muscle concentration in beef cattle (Fig. 6A) and milk concentrations in dairy cows (Fig. 6B). The complete results of sensitivity analyses for all PFAS compounds in all tissues and milk can be found in Supplementary Tables S5–S6. The results showed that AUCs of the muscle concentrations of PFOS and PFHxS in beef cattle were highly sensitive to BW, VLC, partition coefficient of liver (PL) and muscle (PM), KbileC, KehcC and KfecesC, while AUCs of the muscle concentration of PFOA were sensitive to the parameters of Free, BW and PM. For dairy cows (Fig. 6B), the sensitive parameters varied by PFAS compounds. The parameters of BW, VLC, PL, Free, KbileC, KehcC, KfecesC and PMilkM were all sensitive on AUCs of milk concentrations in PFOS, while the BW, Free and PMilkM were sensitive on AUCs of milk concentration in PFHxS. In addition, the parameters BW, KfecesC and PMilkM were highly sensitive parameters on AUCs of PFOA.

3.4. Population PBPK model and its application in the withdrawal interval estimation

The population PBPK model with Monte Carlo simulations was used to estimate WDIs for selected PFAS compounds in beef cattle and dairy cows based on low, medium, and high exposure dose levels for an

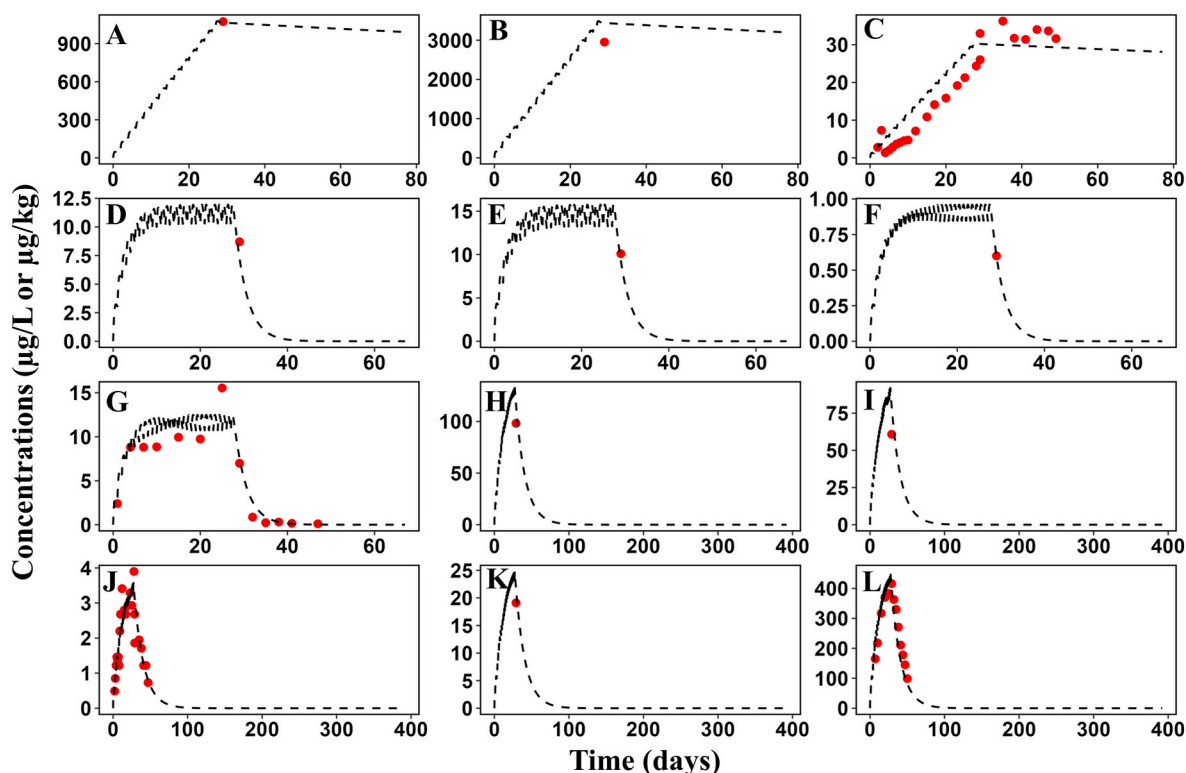


Fig. 4. Model calibration results for individual datasets of the PBPK model for PFOA, PFOS and PFHxS in dairy cows. Comparisons of model predictions (dashed lines) and observed data (red circles) are shown for concentrations of PFOA, PFOS and PFHxS in the plasma, liver, kidney, muscle and milk from dairy cows. All experimental data (A–L) are from the study of [Kowalczyk et al. \(2013\)](#). Plots (A, B, C) represent PFOS concentrations in kidney, liver, and milk, respectively; (D, E, F, G) represent PFOA concentrations in kidney, liver, muscle, and plasma, respectively; (H, I, J, K, L) represent PFHxS concentrations in kidney, liver, muscle, and plasma, respectively. Refer to [Table 1](#) for the detailed dosing regimens from respective experiments for each of the PFAS compounds. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

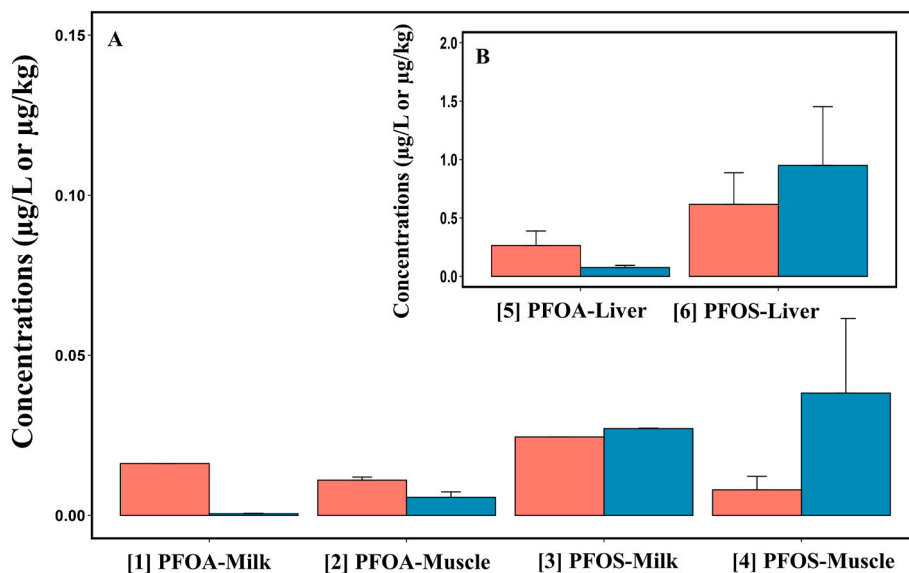


Fig. 5. Evaluation results of the PBPK model for PFOA, PFOS and PFHxS in beef cattle and dairy cows. The red and green bars represent the observed and predicted values, respectively. Experimental studies of [1 and 3] represent PFOA and PFOS concentrations in milk for dairy cows obtained from [Xing et al. \(2016\)](#). Experimental studies of [2, 4, 5, and 6] represent PFOA and PFOS concentrations in muscle and liver for beef cattle obtained from [Wang et al. \(2017\)](#). Because there is no exposure information available, the PFOA and PFOS exposure doses in beef cattle (PFOA: 5.19×10^{-6} mg/kg/day; PFOS: 6.26×10^{-7} mg/kg/day) and dairy cows (PFOA: 5.88×10^{-6} mg/kg/day; PFOS: 7.63×10^{-7} mg/kg/day) were calculated based on [Equations S16–S17](#) and the exposure duration was assumed to be 2 years. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

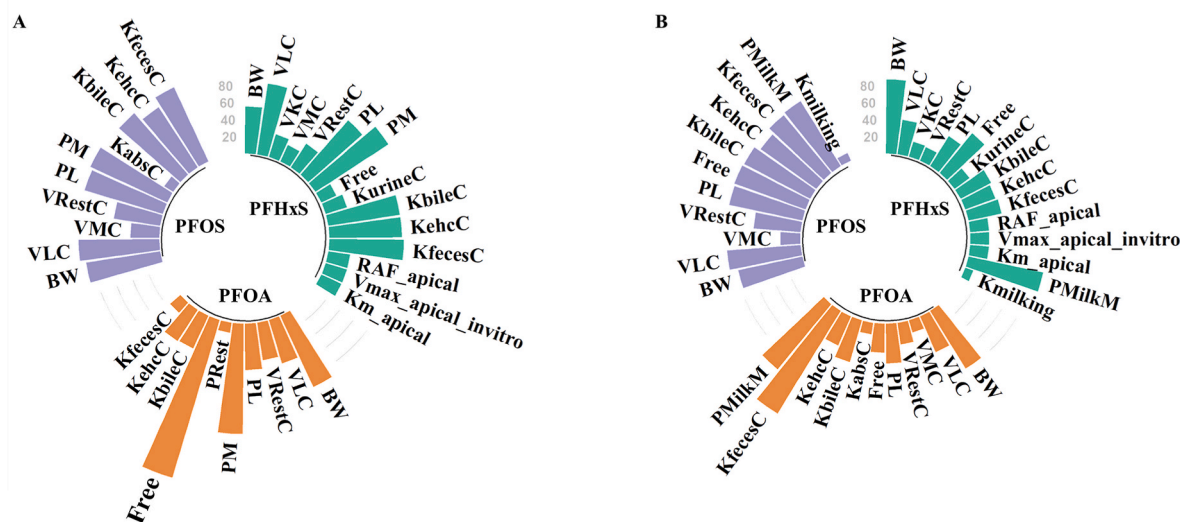


Fig. 6. Normalized sensitivity coefficients (NSCs) of post-optimized parameters using area under curves (AUCs) of PFOA, PFOS and PFHxS in (A) muscle for beef cattle and (B) milk for dairy cow. Only parameters with at least one absolute value of NSC >10% are shown on the plots. Refer to [Tables 2–4](#) for definitions of parameters.

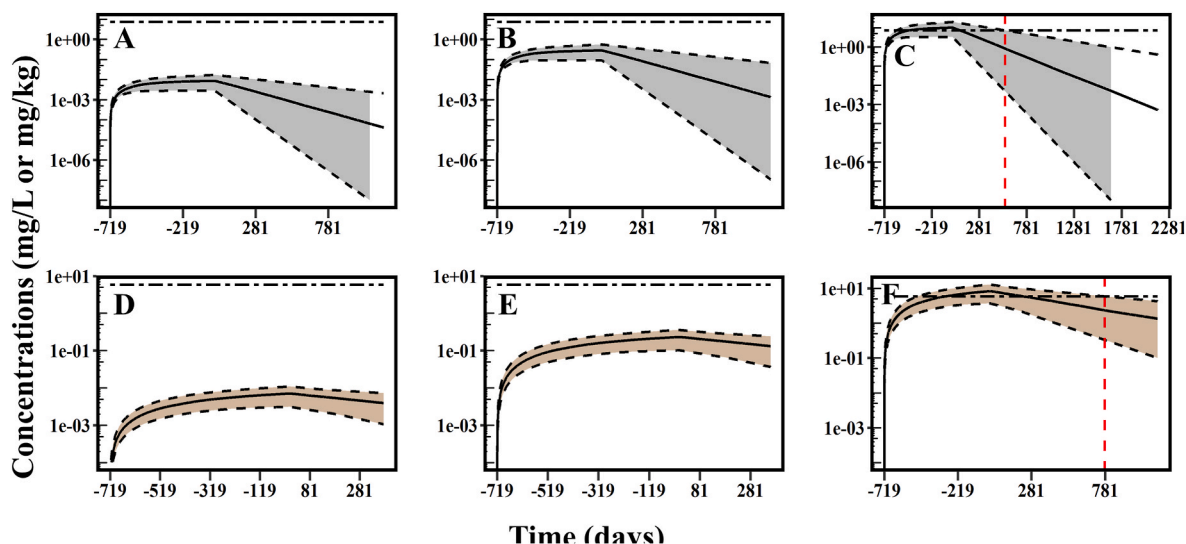


Fig. 7. Model Carlo simulation results for PFOS in muscle for beef cattle (A, B, and C) and milk for dairy cows (D, E, and F) based on low (cattle: 1.65×10^{-7} mg/kg/day; dairy cows: 2.45×10^{-7} mg/kg/day), medium (cattle: 5.36×10^{-6} mg/kg/day; dairy cows: 7.98×10^{-6} mg/kg/day) and high (cattle: 1.93×10^{-4} mg/kg/day; dairy cows: 2.88×10^{-4} mg/kg/day) exposure scenarios, respectively. Each of the simulations was run for 1,000 iterations. The median (black solid line), 1st, and 99th (black dashed lines) percentiles of simulated results were plotted. Tolerance (TOL) for PFOS (beef cattle: 10 mg/kg in muscle; dairy cows: 5.9 ng/mL) is shown in the plot using a black horizontal dotted line. The intersection between the x-axis and the red vertical line indicates the estimated withdrawal interval. Refer to [Table S4](#) for the details about tolerances (e.g., action levels) of different compounds in different tissues in both beef cattle and dairy cows. Both exposure scenarios assumed that beef cattle and dairy cows were exposed to PFOS-contaminated water and forage for 2 years. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

exposure duration of 2 years ([Fig. 7](#) and [Table 5](#)). The low exposure dose levels were 0.187 ng/kg/day in cattle and 0.278 ng/kg/day in dairy cows for PFOA; 0.165 ng/kg/day in cattle and 0.245 ng/kg/day in dairy cows for PFOS; 0.356 ng/kg/day in cattle and 0.53 ng/kg/day in dairy cows for PFHxS. The medium exposure dose levels were 1.35 ng/kg/day in cattle and 2.01 ng/kg/day in dairy cows for PFOA; 5.36 ng/kg/day in cattle and 7.98 ng/kg/day in dairy cows for PFOS; 1.97 ng/kg/day in cattle and 2.94 ng/kg/day in dairy cows for PFHxS. The high exposure doses were 49.3 ng/kg/day in cattle and 73.4 ng/kg/day in dairy cows for PFOA; 193 ng/kg/day in cattle and 288 ng/kg/day in dairy cows for PFOS; 161 ng/kg/day in cattle and 240 ng/kg/day in dairy cows for PFHxS. The time-course PFOS concentrations in muscle of cattle were

lower than the calculated operational tolerance (10 ng/g) following low ([Fig. 7A](#)) and medium ([Fig. 7B](#)) exposure scenarios, while the WDI of PFOS in muscle of cattle based on the high exposure scenario was estimated to be 449 days ([Fig. 7C](#)). Similarly, PFOS concentrations in milk from dairy cows were both lower than the calculated safety concentration (5.9 μ g/L) in the low ([Fig. 7D](#)) and medium exposure scenarios ([Fig. 7E](#)), yet PFOS concentrations in milk of dairy cows in the high exposure scenario required 1,345 days to fall below the calculated safety concentration after last exposure ([Fig. 7F](#)). The WDIs of muscle, liver, kidney in beef cattle and dairy cows were also estimated using the population PBPK model ([Table 5](#)). The longest WDIs among tissues and PFAS compounds in cattle and dairy cows after different exposure

Table 5

Comparisons of the estimated withdrawal intervals (days) for perfluorooctanoic acid (PFOA), perfluorooctane sulfonate (PFOS) and perfluorohexane sulfonate (PFHxS) for beef cattle and dairy cows based on environmental low, medium, and high exposure scenarios.

	PFOA			PFOS			PFHxS		
	Low ^a	Med	High	Low	Med	High	Low	Med	High
Beef Cattle									
Liver	NA ^b	NA	NA	NA	NA	972	NA	NA	16
Muscle	NA	NA	NA	NA	NA	449	NA	NA	69
Kidney	NA	NA	NA	NA	NA	NA	NA	NA	NA
Dairy cows									
Liver	NA	NA	NA	NA	NA	3,671	NA	NA	14
Muscle	NA	NA	NA	NA	NA	2,240	NA	NA	36
Kidney	NA	NA	NA	NA	NA	NA	NA	NA	NA
Milk	NA	NA	NA	NA	NA	1,345	NA	NA	NA

Note.

^a Exposure scenarios were estimated based on Equations S1–S4 and Tables S1–S7. Low, Med and High represent the low- (PFOA: 1.87e-7 mg/kg/day for beef cattle and 2.78e-7 mg/kg/day for dairy cows; PFOS: 1.65e-7 mg/kg/day for beef cattle and 2.45e-7 mg/kg/day for dairy cows; PFHxS: 3.56e-7 mg/kg/day for cattle and 5.30e-7 mg/kg/day for dairy cows), medium- (PFOA: 1.35e-6 mg/kg/day for beef cattle and 2.01e-6 mg/kg/day for dairy cows; PFOS: 5.36e-6 mg/kg/day for beef cattle and 7.98e-6 mg/kg/day for dairy cows; PFHxS: 1.97e-6 mg/kg/day for beef cattle and 2.94e-6 mg/kg/day for dairy cows) and high-exposure scenarios (PFOA: 4.93e-5 mg/kg/day for beef cattle and 7.34e-5 mg/kg/day for dairy cows; PFOS: 1.93e-4 mg/kg/day for beef cattle and 2.88e-4 mg/kg/day for dairy cows; PFHxS: 1.61e-4 mg/kg/day for beef cattle and 2.40e-4 mg/kg/day for dairy cows), respectively.

^b NA indicates that PFAS concentrations in edible tissues or milk were below the tolerance, and thus there was no withdrawal interval needed (i.e., practical zero withdrawal). Note that when the predicted withdrawal interval is zero day, it is common to recommend one day because this is extra-label exposure.

scenarios were chosen to be the model derived WDIs. The estimated WDIs for PFOS and PFHxS following high exposure scenarios (PFOS: 193 ng/kg/day for beef cattle and 288 ng/kg/day for dairy cows; PFHxS: 161 ng/kg/day for beef cattle and 240 ng/kg/day for dairy cows) were 972 days (liver) and 449 days (muscle) for beef cattle, and 3,671 days (liver) and 2,240 days (muscle) for dairy cows. For low and medium exposure scenarios, the PFAS levels were lower than the operational tolerance in edible tissues or in milk for beef cattle and dairy cows, respectively.

3.5. Web-based interactive PBPK visualization platform

Fig. 8 shows a representative screenshot of the developed web-based

PBPK interface of the population PBPK model that can be used to estimate WDIs of PFOA, PFOS, PFHxS in beef cattle and dairy cows. This web-based interface was designed with RShiny and has been published online via the following link: https://pbpk.shinyapps.io/igPFAS_PBPK/. Readers and users can input the parameter values into the interface and run a Monte Carlo simulation with 1,000 iterations to calculate PFOA, PFOS and PFHxS concentrations in plasma, muscle, liver, kidney, fat and milk based on different exposure scenarios in beef cattle and dairy cows. In addition, a detailed report of the simulation result will be generated and can be downloaded, which provides the estimated WDI for selected PFAS compounds with the selected set of parameters and exposure scenario in the chosen species. The WDI plot allows users to find the exact time when the selected PFAS compound is below the tolerance or

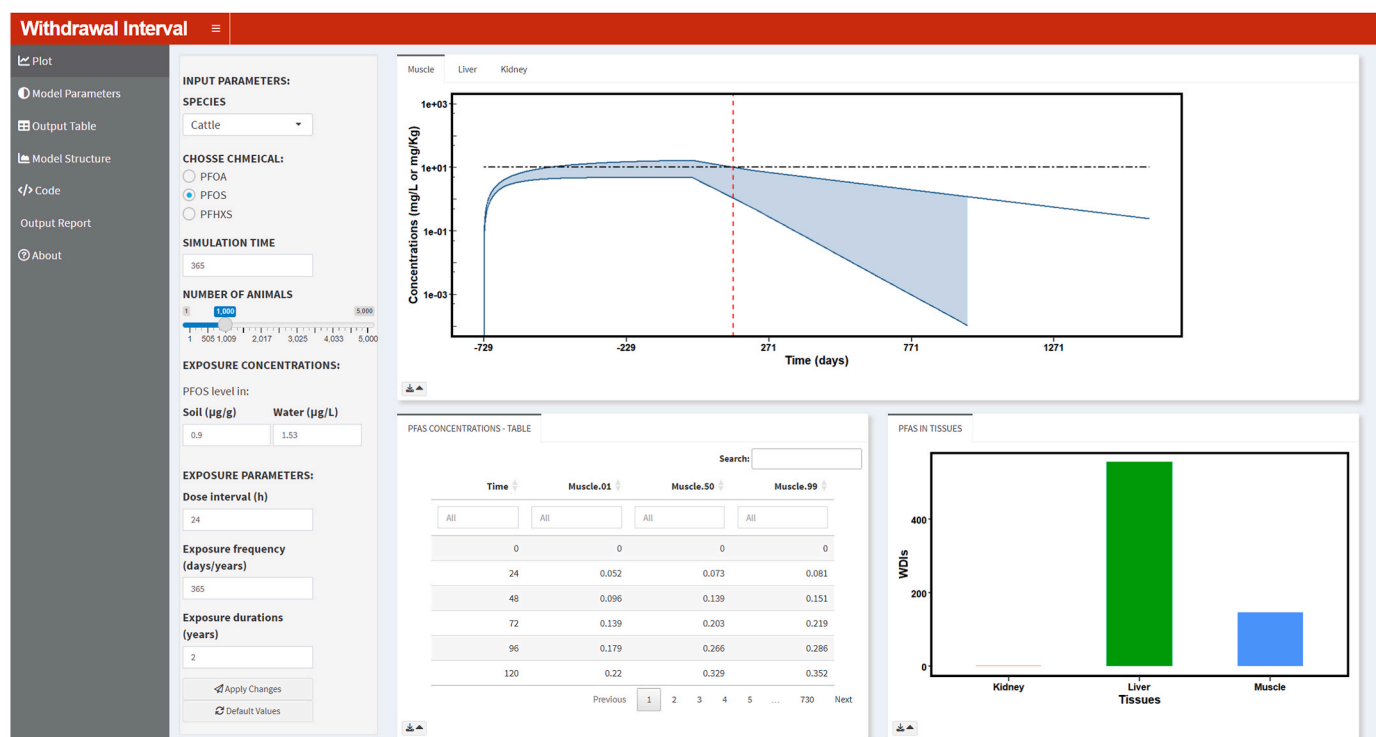


Fig. 8. A representative screenshot of the user-friendly interactive generic PBPK (igPBPK) interface for PFOA, PFOS, and PFHxS for beef cattle and dairy cows.

action level in a specific tissue or milk.

4. Discussion

This study established a web-based PBPK model platform that is capable of simulating plasma, tissue, and milk residues, and estimating WDIs of several PFAS compounds in beef cattle and dairy cows. Three commonly detected long-chain PFAS chemicals, namely PFOA, PFOS and PFHxS, were tested in this study. In this web-based PBPK model platform, population-based simulations can be generated by using Monte Carlo simulations to estimate WDIs of edible tissues for the three selected PFAS compounds following different customized exposure scenarios. Our results suggest that the WDIs can vary widely between PFAS compounds, target tissues, regulatory standards (i.e., tolerances) and exposure scenarios; highlighting the importance of using physiologically-based mechanistic models to infer the influences of these variables on the residues in edible tissues and provide a scientifically-based WDI. This PBPK interface serves as a powerful and user-friendly tool to ensure the animal-derived food products are devoid of violative PFAS residues and thereby pose a minimal risk to human health. This PBPK model platform can be extended to other PFAS compounds and to other food animal species to help address food safety issues related to PFAS contamination in animal-derived food products for the U.S. and other countries.

Several PBPK or toxicokinetic models for specific PFAS compounds (e.g., PFOA, PFOS, PFHxS and PFDA) have been developed to aid risk assessment (Andersen et al., 2006; Chou and Lin, 2019, 2021; Loccisano et al., 2011, 2012a, 2012b; Tan et al., 2008; Wambaugh et al., 2013; Worley and Fisher, 2015; Worley et al., 2017). Some of these existing models were able to be applied to multiple experimental animal species (Cheng and Ng, 2017; Chou and Lin, 2019; Loccisano et al., 2012b) and further extrapolated to humans (Brochot et al., 2019; Chou and Lin, 2021; Deepika et al., 2021). However, existing PFAS PBPK models are mostly limited to one or two specific PFAS compounds, and none of these were developed for food-producing animals to address food safety issues, despite the fact that one of the most significant sources of human exposure to PFAS is dietary intake *via* animal-derived food products (Death et al., 2021; Sunderland et al., 2019). Additionally, there is neither a reliable scientific approach to know how much PFAS is transferred from contaminated feed or water to food-producing animals or animal products, nor are there federal food safety standards for PFAS concentrations in animal-derived food products. In this study, we developed a PBPK model for several PFAS compounds and converted it to a user-friendly web interface, which allows users without programming experience to conduct PBPK modeling simulations. By incorporating multimedia exposure equations into the model, the exposure scenarios/dosing regimens can be customized and estimated by users. This is particularly useful for food-producing animals exposed to PFAS-contaminated feed or water in the farm as the users can estimate the PFAS concentrations of edible tissues in livestock based on the detected PFAS levels in soil and water. We expect that the concept of an interactive generic PBPK model platform for PFAS in food-producing animals can be used as a science-based risk assessment tool to accelerate the establishment of regulatory standards for PFAS in animal-derived food products.

It is important to note that the PBPK-estimated WDIs vary widely between different PFAS compounds, in different tissues, and among different exposure scenarios. Our results suggest that the WDI of PFOS is much longer than other PFAS compounds such as PFOA and PFHxS in cattle and cows, which is consistent with previous studies (Lupton et al., 2012, 2014). A series of experiments conducted by USDA Food Safety and Inspection Services (FSIS) with PFOA and PFOS in beef cattle reported the elimination half-lives were significantly different between PFOA and PFOS in beef cattle (Lupton et al., 2012, 2014). The majority of PFOA was excreted in urine within 9 days with a plasma elimination half-life of 19.2 h when Angus cattle were administered a single oral

dose of 1.0 mg/kg BW of ^{14}C -PFOA (Lupton et al., 2012). In contrast to PFOA, PFOS showed a different pharmacokinetic behavior in the body of cattle when given a single oral dose of 8 mg/kg BW (Lupton et al., 2014). PFOS was found to be primarily stored in plasma, liver, bile, and kidney of Angus steers and was excreted through the feces. The half-life of PFOS in plasma was estimated to be 116 days (Lupton et al., 2015), which is much longer than those observed for PFOA (Lupton et al., 2012). In other studies conducted in Australia (Drew et al., 2021, 2022), the serum half-life of PFHxS was estimated to be about 9 days, which is closer to, but still much longer than the serum half-life of PFOA estimated in the Lupton et al.'s study (Lupton et al., 2012). Based on the WDI calculations in this study (Table 5), the longest WDIs were found in liver for PFOS (cattle: 972 days; cows: 3,671) and muscle for PFHxS (cattle: 69 days; cows: 36) following the high exposure scenario, yet PFOA concentrations in edible tissues were below the calculated safe concentration for both exposure scenarios. Compared with beef cattle (muscle: 449 days; liver: 972 days), estimated WDIs of PFOS for dairy cows (muscle: 2,240 days; liver: 3,671 days) were much longer. The exact reason for this discrepancy is unknown. However, it might be because dairy cows have a higher percentage of fat that serves as an accumulation depot for PFAS, thus PFAS compounds have a larger volume of distribution and slower elimination in dairy cows than in beef cattle. Tissue distribution and excretion of PFAS appear to differ between species (Death et al., 2021; Guruge et al., 2008). Differences in elimination half-lives between species have been reported in non-farm animals (Chang et al., 2012; Lou et al., 2009; Seacat et al., 2002), but relatively few studies reported the pharmacokinetic differences of PFAS between different food-producing animal species and between different production classes (i.e., primarily different age groups) in the same farm animal species (Death et al., 2021; Guruge et al., 2008).

There are several limitations to this study. First, our PBPK model was developed based on the previous rat model, thus this extrapolation from monogastric species to ruminants adds uncertainty due to the differences in oral digestion/absorption between two species. Ruminants are unique in their mode of oral plant digestion because they evolve a forestomach which allows microbial digestion and fermentation of celluloses (Bergman, 1990), thereby the biological mechanisms of absorption and transport of chemicals might be different between ruminants and monogastric species (Riviere, 2011). However, the detailed biological mechanisms underlying the oral absorption of PFAS compounds in ruminants are still unclear (Lupton et al., 2012). Further studies are needed to characterize the absorption and transport of chemicals, microbial digestions and transport through the entire bovine four-compartment stomach. Despite these limitations, it must be stressed that the PBPK model developed in this study was parameterized with bovine physiological values and the model adequately predicted the PFAS disposition in several independent cattle studies, suggesting that these factors are important but not essential for modeling these compounds. Second, although our PBPK model has incorporated multimedia exposure equations to estimate the PFAS exposure to food-producing animals based on PFAS concentrations in water and soil, the model does not include additional exposure sources such as dust, groundwater, and pasture. In addition, only three selected PFAS compounds were included in the present PBPK model, thus the predictions of WDIs were limited to PFOA, PFOS and PFHxS, despite the fact that farm animals might be exposed to multiple other PFAS compounds. Third, during model development, precise exposure information (e.g., dosing levels and exposure durations) was needed to calibrate the model with the observed pharmacokinetic data from farm animals. However, exposure dose and duration are difficult to obtain when these data were collected from the literature where livestock were fed with naturally contaminated feed, introducing uncertainty and variability, although the simulation results adequately corresponded to the observed data obtained from different studies that were used to parameterize the model. Fourth, a significant variability in PFAS concentrations in tissues and WDIs within and between different production groups (i.e., beef

cattle versus dairy cows) have been noted in this study. More comprehensive studies to characterize the inter- and intra-species differences in the pharmacokinetics of PFAS compounds between species of farmed animals and between different age groups of the same farm animal species are warranted. In our previous study, Bayesian Markov chain Monte Carlo simulations were incorporated into a PBPK model to quantitatively characterize the variability of pharmacokinetic parameters between rodents, monkeys and humans (Chou and Lin, 2019). Based on this approach, the present PBPK modeling platform can be further optimized to characterize the variability of pharmacokinetics between beef cattle and dairy cows. Finally, the current model can be used to predict the WDIs for the selected three PFAS compounds individually, but it cannot predict WDIs when these three PFAS compounds are consumed concurrently because the model does not account for the potential pharmacokinetic interactions of PFAS mixtures.

5. Conclusions

This manuscript reports a new generic PBPK model for three PFAS compounds in beef cattle and dairy cows. The model was calibrated and evaluated with published pharmacokinetic data for the three most commonly detected PFAS compounds (i.e., PFOA, PFOS, PFHxS) in human plasma and in the environment. The model simulation results adequately captured the observed pharmacokinetic data for selected PFAS compounds in plasma, edible tissues, and milk of beef cattle and dairy cows. The model was applied to predict tissue residues and WDIs of selected PFAS compounds following low, medium, and high environmental exposure scenarios, respectively. The results showed that the predicted tissue residues and WDIs are highly dependent on the specific PFAS chemical's toxicokinetic properties and exposure levels. For PFOA, no withdrawal (zero hours) is needed following low, medium, and high environmental exposures. For PFOS and PFHxS, there is no concern following low and medium environmental exposures, but it may take up to 3 years for beef cattle and up to 10 years in dairy cows for PFOS to deplete to the calculated safe concentrations following a high environmental exposure scenario. In many cases, this exceeds the animal's production life span. The final model was converted to a web-based user-friendly visualization platform to facilitate applications of our PBPK model in the estimation of chemical residues and WDIs in edible tissues or milk based on the detected water or soil sample concentrations of PFAS compounds on farms. Despite some limitations remaining, this modeling framework provides a basis to apply the next generation PBPK models to support food safety assessment and it also provides a robust basis to extrapolate to other PFAS compounds and other food-producing animals, such as sheep and goats.

CRediT authorship contribution statement

Wei-Chun Chou: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Lisa A. Tell:** Conceptualization, Funding acquisition, Writing – review & editing. **Ronald E. Baynes:** Funding acquisition, Writing – review & editing. **Jennifer L. Davis:** Funding acquisition, Writing – review & editing. **Yi-Hsien Cheng:** Investigation, Data curation, Writing – review & editing. **Fiona P. Maunsell:** Funding acquisition, Writing – review & editing. **Jim E. Riviere:** Conceptualization, Funding acquisition, Writing – review & editing. **Zhoumeng Lin:** Conceptualization, Methodology, Software, Validation, Resources, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have provided the data and code in this GitHub repository (<https://github.com/UFPBPK/FARAD-PFAS>).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fct.2023.114062>.

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